

Retail Inventory & Supply Chain Analytics

A business analytics project exploring inventory health, stockout risk, supplier performance, and profitability across a simulated multi-store retail operation.

PostgreSQL

Power BI

DAX

Excel

Project Overview

What this project is about and what it set out to answer

The Context

Simulated a multi-store retail grocery dataset modelled on how a business like FoodCo operates. Five stores, six product categories, six suppliers.

The Objective

Use real data analysis to find where the business is losing money through stockouts, expiry waste, over-ordering, and supplier problems.

The Deliverable

SQL business analysis in PostgreSQL and a four-page interactive Power BI dashboard built to drive real operational decisions.

5 Stores

6 Product Categories

6 Suppliers

7 Business Analyses

4 Dashboard Pages

The Business Problems

Six real operational challenges this project was built to address

"A customer walks in; the shelf is empty. That sale is gone. Meanwhile in the stockroom, 300 products are quietly expiring."

01

High Stockout Rate

38.2% of products go out of stock regularly, costing the business lost sales every single week.

02

Expiry Losses

154 products already expired. 301 more expiring in 30 days. Mostly Dairy and Frozen Foods.

03

Over-Ordering

Some products restocked at 5x their sales rate, wasting money and filling storage unnecessarily.

04

Supplier Damage

Fragile products arriving damaged, creating unexpected gaps on shelves with no warning.

05

Critical Inventory

458 products at critical stock levels with only 1 to 4 days of coverage remaining.

06

Weak Promotions

Promotions are running across all categories but generating very little additional revenue.

Dataset & Methodology

How the data was structured and how the analysis was built

DATA TABLES

products

Product names, categories, and pricing

stores

Five store locations and identifiers

suppliers

Supplier info and lead time data

inventory_supply_chain

Main fact table: stock, sales, expiry, replenishment, damage

ANALYTICAL WORKFLOW

1

Raw CSV Files

Four datasets prepared and staged in Excel

2

PostgreSQL

Schema design, data loading, and cleaning

3

7 Business Queries

SQL analysis across seven operational areas

4

SQL Analytical View

Single clean reporting dataset for Power BI

5

Power BI + DAX

Four-page interactive dashboard with KPI measures

Sales & Profitability

Which products, categories, and stores are actually driving the business

N54M

Total Revenue

N14M

Total Profit

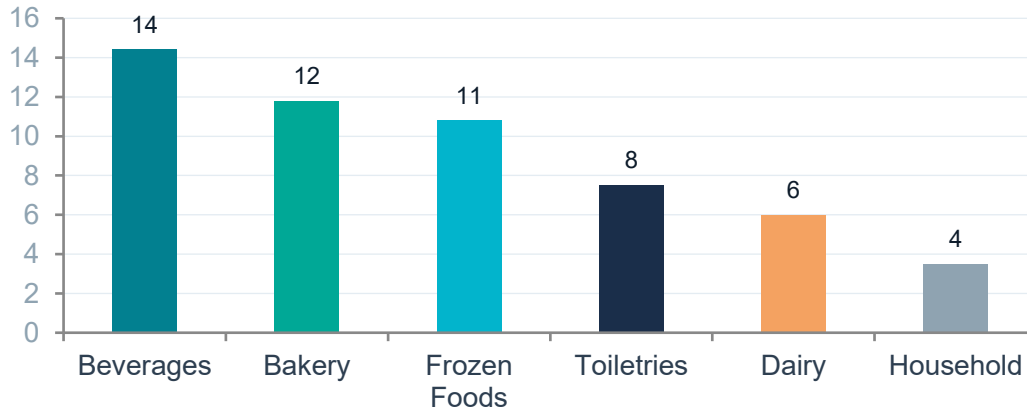
26.6%

Profit Margin

Bread

Top Product
(N1.48M Revenue)

Revenue by Product Category (NM)



Beverages leads all categories

N14.4M in revenue. Coca-Cola and Maltina are the main drivers within this category.

Bread is the top single product

N1.48M in revenue across all stores. A stockout on Bread hits the business harder than almost anything else.

All 5 stores perform closely

FoodCo Mokola leads at N11.25M but all stores are within a narrow range, showing consistent operations.

Margins are stable across categories

Profit margins range from 26.5% to 26.8%. No single category is significantly more profitable than another.

Inventory Health

How much stock is actually available and how many days before shelves go empty

458

Critical Stock
Products

384

Low Stock
Products

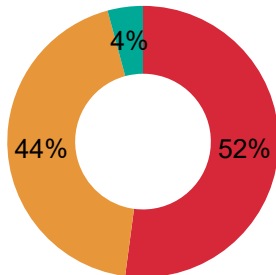
37

Healthy Stock
Products

1-4

Days of Coverage
(Fast Movers)

Inventory Health Distribution



■ Critical (458) ■ Low Stock (384) ■ Healthy (37)

95% of inventory is at risk

Only 37 product records are actually healthy. The entire inventory is running on thin margins with almost no buffer.

Fast movers have 1-4 days left

Products like Bread and Coca-Cola have critically low coverage. One supplier delay and shelves go empty.

Reorder points are not working

Products are repeatedly falling below reorder thresholds, especially those with stronger demand. The system needs a reset.

Safety stock is too low

High-demand products need a meaningful inventory buffer to absorb supply disruptions before they cause a stockout.

Finding: The business is one supply disruption away from widespread empty shelves across all five stores.

Stockout Risk

Which products and stores go out of stock most, and what is causing it

38.2%

Overall Stockout
Rate

40.8%

Worst Store
(FoodCo Jericho)

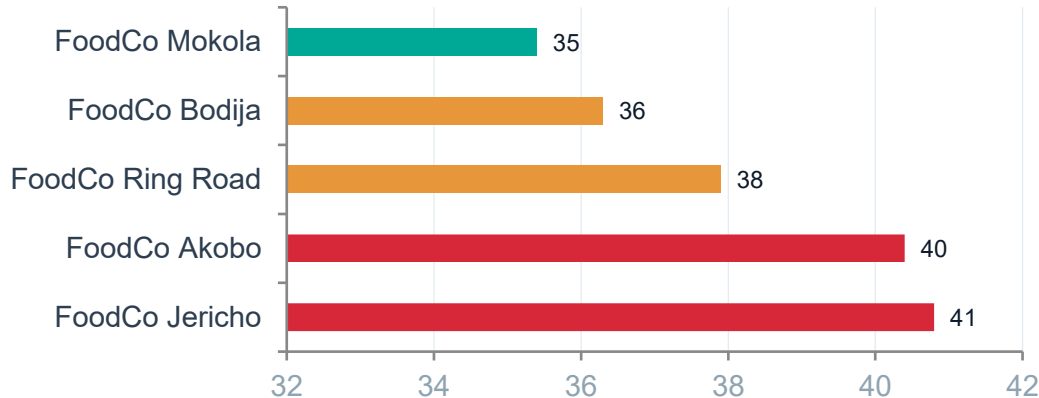
100%

Biscuits Stockout
Rate

35.4%

Best Store
(FoodCo Mokola)

Stockout Rate by Store (%)



Biscuits never in stock

A 100% stockout rate means Biscuits was out of stock every single time it was checked. Bread and Butter were also frequent offenders.

Stockouts happen in low demand too

This is the most important finding. Shortages are not caused by demand spikes. They are caused by poor replenishment planning.

Jericho and Akobo need attention

Both stores exceed 40% stockout rates. This points to either ordering behaviour or supplier reliability issues at those locations.

Bakery and Dairy most vulnerable

These categories experience the highest stockout frequency and also happen to be among the top revenue drivers.

Finding: Stockouts are a planning failure, not a demand problem. Products run out even when customers are not buying much.

Expiry & Inventory Waste

Products expiring before they sell and the categories carrying the most risk

154

Already Expired
Products

301

Expiring Within
30 Days

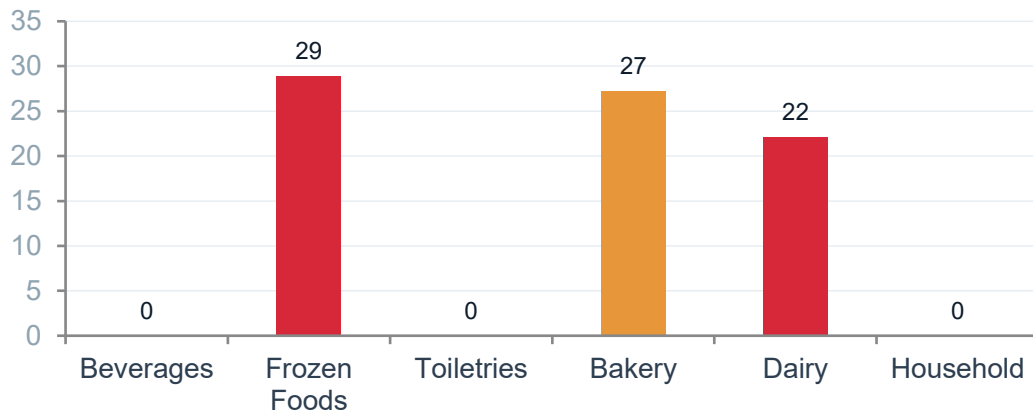
455

Total Products
at Expiry Risk

Dairy

Highest Expiry
Risk Category

Expired Products % by Category



Dairy and Frozen Foods at highest risk

Butter, Cheese, Yoghurt, Frozen Fish, and Ice Cream make up the bulk of expired and near-expiry inventory.

Over-ordering is making it worse

The same products expiring are also being over-ordered. Stock arrives faster than it can be sold, guaranteeing future losses.

Every expired unit is a direct loss

There is no recovering that cost. Unlike a stockout which is a missed sale, expiry is money already spent and thrown away.

Near-expiry stock needs promotions

Targeted discounts before the expiry date can turn a guaranteed loss into a partial recovery. This is a simple operational win.

Finding: The categories losing the most to expiry are the same ones being over-ordered. Better planning solves both problems at once.

Supplier Performance

Delivery reliability and damaged goods across all six suppliers

6

Total Suppliers

4.70

Avg Lead Time
(Days)

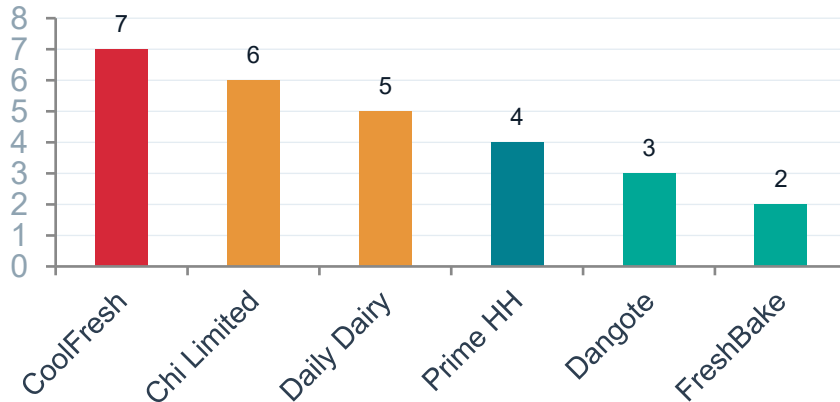
3.99

Avg Damaged
Units Per Order

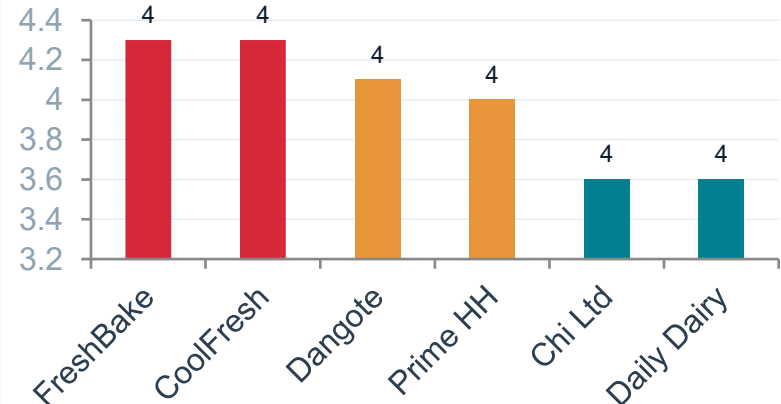
7 Days

Longest Lead Time
CoolFresh Logistics

Supplier Lead Time (Days)



Supplier Damage Rate (Avg Units)



Finding: Delivery timing is acceptable. The real risk is damage rates, especially for frozen and fragile products from FreshBake and CoolFresh.

Replenishment Efficiency

Are products being restocked in the right quantities relative to what is actually being sold

555

Overstocked
Products

2.08x

Avg Restock-to-
Sales Ratio

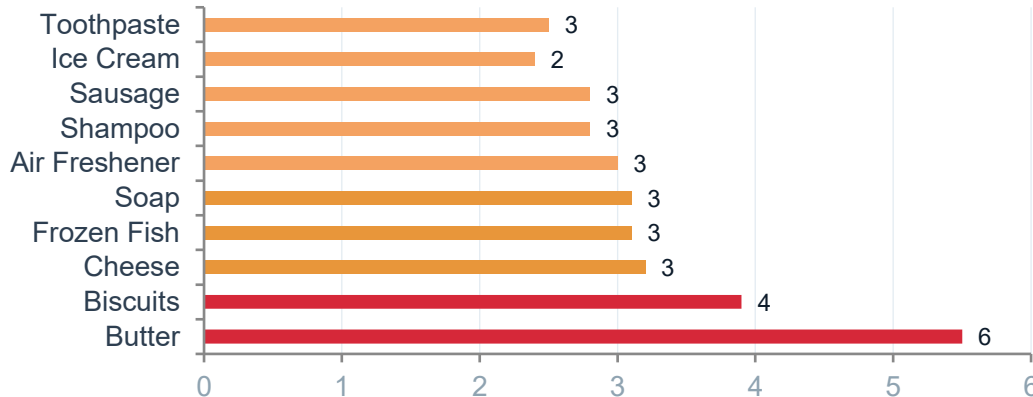
5.5x

Worst Offender
Butter

Dairy

Most Overstocked
Category

Most Over-Ordered Products (Restock Divided by Sales)



Butter restocked 5.5x its sales

For every unit sold, 5.5 are ordered. Stock piles up faster than it moves, directly feeding into expiry losses.

Frozen Foods worst overall

Frozen Foods category average is 80.99 units restocked vs only 30.28 units sold. Nearly 3x over-ordered across the board.

Over-ordering is a cash problem

Money tied up in excess stock cannot be used elsewhere. It also inflates storage costs and increases expiry exposure.

Replenishment needs sales anchoring

Order quantities must be based on actual recent sales data, reviewed on a rolling 30-day basis, not fixed estimates.

Finding: Products are being ordered at up to 5x their actual sales rate. The business is spending money on stock it cannot sell.

Promotion Effectiveness

Are promotions actually moving more products or just running in the background

-6.70%

Overall Promo
Impact on Revenue

-8.7%

Promo Impact
vs Prior Month

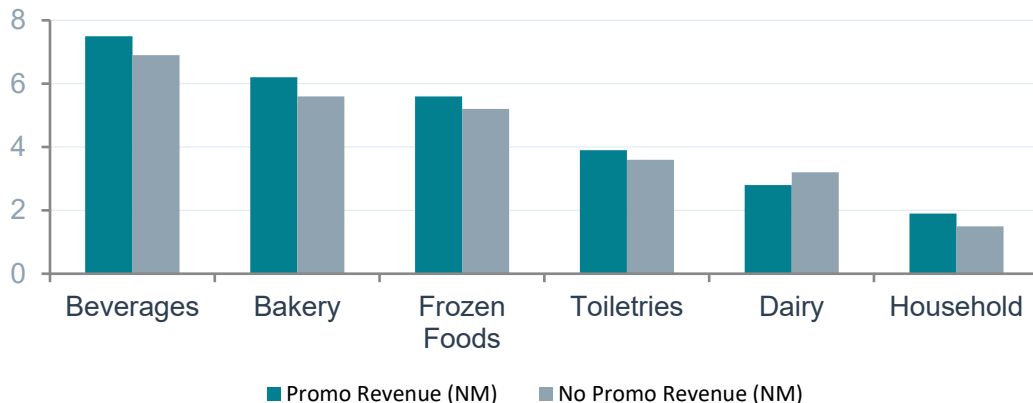
Dairy

Best Responding
Category

Household

Weakest Responding
Category

Promo vs No-Promo Revenue by Category (NM)



Promotions are not working well

The revenue gap between promotional and non-promotional periods is small across nearly every category. The spend is not justified by the return.

Dairy responds slightly better

Of all categories, Dairy shows the most meaningful difference between promo and no-promo revenue. This is where promotion effort should focus.

Stockouts happen during promos too

Running promotions on products that are out of stock is a waste. Inventory planning must be coordinated with any promotional activity.

Strategy needs to change

Broad promotions running across all categories should be replaced with targeted, time-limited campaigns in categories that actually respond.

Finding: Current promotions generate very little additional revenue. A more targeted approach focused on responsive categories is needed.

Power BI Dashboard Overview

Four pages, each answering one clear business question a manager would ask

01

Executive Overview

How is the business performing right now?

Revenue, profit, store comparison, and category contribution. Built for leadership to get a quick health check before diving into specifics.

02

Inventory Health & Risk

Where are the gaps and what needs action today?

Critical stock flags, expiry risk by category, stockout rates by store, and a live restock action table showing OK, LOW STOCK, or REORDER.

03

Supplier & Replenishment

Are suppliers reliable and are we ordering the right amounts?

Lead time comparison, damage rates per supplier, over-ordering patterns by product, and restocking efficiency across all six categories.

04

Profitability & Promotions

What is actually driving our margins?

Profit margins by category, top revenue products, monthly profit trends across 12 months, and promotional impact compared to non-promotional periods.

Key Business Outcomes

Six clear findings a supply chain team can act on immediately

Inventory is dangerously thin

95% of products are at critical or low stock. A single supply delay could cause widespread shortages across all five stores at the same time.

Stockouts are a planning failure

38.2% stockout rate during low-demand periods proves this is not a customer behaviour problem. It is a replenishment strategy problem with a clear solution.

Perishables are a double loss

Dairy and Frozen Foods are simultaneously being over-ordered and expiring. The business is paying for stock it is then throwing away.

Supplier damage needs tracking

FreshBake and CoolFresh have the highest damage rates. Without consistent data, there is no leverage to demand better handling from suppliers.

Few products carry most revenue

Bread, Coca-Cola, Meat Pie, and Maltina carry a disproportionate share of total revenue. A stockout on any of these hits the business harder than anything else.

Promotions need rethinking

Current promotional activity is broad and generates very little return. Focusing on categories that actually respond would produce better results with less spend.

Recommendations

Six practical actions with clear priority levels the business can start implementing

01 Reset All Reorder Points

HIGH

Review and rebuild reorder thresholds for every product based on actual 30-day rolling sales velocity. Stop relying on fixed estimates.

03 Cut Perishable Order Quantities

HIGH

Reduce replenishment quantities for Dairy and Frozen Foods immediately. Use recent sales data to recalibrate order sizes across both categories.

05 Introduce Supplier Damage Tracking

MEDIUM

Log damaged arrivals consistently for every supplier and product. Use this data in quarterly reviews to hold suppliers to a measurable standard.

02 Protect Top Revenue Products

HIGH

Bread, Coca-Cola, Meat Pie, and Maltina must have dedicated safety stock buffers. These cannot go out of stock under any circumstances.

04 Run Near-Expiry Promotions

MEDIUM

Before products cross their expiry date, use targeted discounts or bundle deals to clear them. A partial recovery is always better than a full write-off.

06 Redesign the Promotions Strategy

LOW

Stop running broad promotions across all categories. Focus on Dairy and Beverages which show the best response. Test shorter, targeted campaigns.

What I Learned

Four honest reflections from building this project end to end

01

Data does not speak for itself

Having numbers is not the same as having answers. I had to understand the business first before I could write a single useful query. The best analysis always starts with the right question.

02

Surface causes can be completely wrong

The stockout problem looked like a demand issue. When I looked deeper, it was a replenishment planning failure. If I had stopped at the surface, the recommendation would have been wrong.

03

Making data realistic is harder than it looks

Getting a simulated dataset to behave like a real business took multiple rounds of adjustment. Uniform patterns and disconnected numbers produce analysis that looks correct but means nothing.

04

A dashboard is for decisions, not displays

I removed more charts than I kept. Every visual had to answer a question someone would actually walk in and ask. If it did not help someone make a decision, it had no place on the slide.

Thank You

This project was built to show that good analysis starts with the right questions, not just the right tools.

7

Business Analyses

4

Dashboard Pages

N54M

Revenue Analyzed

6

Suppliers Reviewed

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